

EDUCATION

Max Planck Institute for Intelligent Systems, Tübingen

Germany

PhD Computer Science; Meta PhD Fellow 2023 (21/3200 applicants)

Feb 2021 – Jan 2026

Thesis: Physics-informed modeling of dynamic humans and their interactions

Grade: Summa cum Laude, Advisor: Prof. Michael Black

Carnegie Mellon University, School of Computer Science

Pittsburgh, USA

Master of Science in Computer Vision (MSCV)

Dec 2018

GPA: 4.15/4.33, Advised by Prof. Kris Kitani

Birla Institute of Technology and Science (BITS), Pilani

Hyderabad, India

Bachelor of Engineering with Honors in Electronics and Communication

July 2016

Engineering, Minor in Finance

GPA: 9.16/10 (top 2% among 1500 students, Merit scholarship recipient)

PUBLICATIONS

- **UniCon3R: Unified Contact-aware 4D Human-Scene Reconstruction from Monocular Video**
T Sur, [S Tripathi](#), N Athanasiou, H L Nguyen, K Xu, M J Black, A Yao. (in review)
- **Towards in-the-wild Egocentric 3D Hand-Object Pose Estimation**
S Bansal, J Zhao, [S Tripathi](#), Z Zhu, M J Black, D Damen. (in review)
- **PICO: Reconstructing 3D People In Contact with Objects** <https://pico.is.tue.mpg.de>
[S Tripathi](#), A Cseke, S Dwivedi, A Lakshmipathy, M J Black, D Tzionas. CVPR 2025
- **InteractVLM: 3D Interaction Reasoning from 2D Foundational Models** <https://interactvml.is.tue.mpg.de/>
S Dwivedi, D Antić, [S Tripathi](#), O Taheri, C Schmid, M J Black, D Tzionas. CVPR 2025
- **HUMOS: Human Motion Model conditioned on Body Shape** <https://carstenepic.github.io/humos>
[S Tripathi](#), O Taheri, C Lassner, M J Black, D Holden, C Stoll. ECCV 2024
- **DECO: Dense Estimation of 3D Human-Scene Contact in the Wild** <https://deco.is.tue.mpg.de>
[S Tripathi](#), A Chatterjee, J Passy, H Yi, D Tzionas, M J Black. ICCV 2023 (oral)
- **EMOTE: Emotional Speech-Driven Animation with Content-Emotion Disentanglement**
R Danecek, K Chhatre, [S Tripathi](#), Y Wen, M Black, T Bolkart. SIGGRAPH Asia 2023 <https://emote.is.tue.mpg.de>
- **3D Human Pose Estimation via Intuitive Physics**
[S Tripathi](#), L Muller, C P Huang, O Taheri, M J Black, D Tzionas. CVPR 2023 <https://ipman.is.tue.mpg.de>
- **BITE: Beyond Priors for Improved Three-D Dog Pose Estimation**
N Rüegg, [S Tripathi](#), K Schindler, M J Black, S Zuffi. CVPR 2023 <https://bite.is.tue.mpg.de>
- **MIME: Human-Aware 3D Scene Generation**
H Yi, C P Huang, [S Tripathi](#), L Hering, J Thies, M J Black. CVPR 2023 <https://mime.is.tue.mpg.de>
- **PERI: Part Aware Emotion Recognition In The Wild**
A Mittel, [S Tripathi](#). ECCVW 2022 <https://cvml.page.link/peri>
- **Occluded Human Mesh Recovery**
R Khirodkar, [S Tripathi](#), K Kitani. CVPR 2022 <https://cvml.page.link/ochmr>
- **AGORA: Avatars in Geography Optimized for Regression Analysis**
P Patel, P C Huang, J Tesch, D T Hoffman, [S Tripathi](#), M J Black. CVPR 2021 <https://cvml.page.link/agora>
- **PoseNet3D: Unsupervised 3D Human Shape and Pose Estimation**
[S Tripathi](#), S Ranade, A Tyagi, A Agarwal. 3DV 2020 (oral) <https://cvml.page.link/pose>
- **Learning to Generate Synthetic Data via Compositing**
[S Tripathi](#), S Chandra, A Agarwal, A Tyagi, J Rehg, V. Chari. CVPR 2019 <https://cvml.page.link/learn>
- **C2F: Coarse-to-fine Vision Control System for Automated Microassembly**
[S Tripathi](#), D Jain, H Sharma. Nanotechnology and Nanoscience Asia, 2018 <https://cvml.page.link/c2f>
- **Sub-cortical morphology and voxel based features for Alzheimer's disease classification**
[S Tripathi](#), SH Nozadi, M Shakeri, S Kadoury. ISBI 2017 <https://cvml.page.link/shape>
- **Deep spectral-based shape features for Alzheimer's Disease classification**
M Shakeri, H Lombaert, [S Tripathi](#), S Kadoury. MICCAI-SESAMI, 2016 <https://cvml.page.link/spec>
- **Three-dimensional pose estimation.**
[S Tripathi](#), S Ranade, A Tyagi, A Agarwal. US Patent 11526697
- **Generation of synthetic image data using three-dimensional models.**
[S Tripathi](#), S Chandra, A Agarwal, A Tyagi, J Rehg, V. Chari. US Patent 10909349
- **Generation of synthetic image data for computer vision models**
[S Tripathi](#), S Chandra, A Agarwal, A Tyagi, J Rehg, V. Chari. US Patent 10860836

PATENTS

PICO: Reconstructing 3D People In Contact with Objects

Jun 2024 – Nov 2025

Advisor: Michael Black, Dimitrios Tzionas

- Collected PICO-db, a dataset of 3D contact correspondences using a novel body-to-object contact transfer approach
- Built PICO to recover humans, objects, and their interactions (HOI) - all in 3D, from just a single internet image.

HUMOS: Human Motion Model conditioned on Body Shape

Jun 2023 – Dec 2023

Advisor: Carsten Stoll, Daniel Holden, Christoph Lassner, Michael Black

Epic Games, San Francisco

- Built HUMOS, a self-supervised human motion model conditioned on body shape – trained with novel cycle consistency, intuitive physics and stability constraints

DECO: Dense Estimation of 3D Human-Scene Contact in the Wild

Nov 2022 – Mar 2023

Advisor: Dimitrios Tzionas, Michael Black

MPI-IS, Tübingen

- Collected DAMON, a large-scale dataset with dense vertex-level 3D contact annotations for in-the-wild human-object interactions
- Trained DECO, a novel regressor that predicts vertex-level 3D contacts on a body from a single RGB image

3D Human Pose Estimation via Intuitive Physics

Dec 2021 – Nov 2022

Advisor: Dimitrios Tzionas, Michael Black

MPI-IS, Tübingen

- Proposed novel biomechanically inspired intuitive physics terms that are simple, differentiable and compatible with parametric body models such as SMPL/SMPLX
- Demonstrated that incorporating differentiable physics improves 3D human pose estimation
- Collected Mocap data with extreme poses to test our approach in challenging scenarios

Occluded Human Mesh Recovery

Aug 2021 – Dec 2021

Advisor: Kris Kitani

CMU

Proposed a novel top-down mesh recovery architecture capable of leveraging image spatial context for handling multi-person occlusion and crowding

AGORA: Avatars in Geography Optimized for Regression Analysis

Aug 2020 – Dec 2020

Advisor: Michael Black

MPI-IS, Tübingen

- Developed a 3D human shape and pose estimation model trained on synthetic data that generalizes to real scenes using various 2D and 3D losses

PoseNet3D: Unsupervised 3D Human Shape and Pose Estimation

Feb 2019 – Nov 2019

Collaborators: Amit Agarwal, Amrith Tyagi

Amazon Lab126

- Proposed self-consistency and adversarial losses to train a novel unsupervised model to estimate 3D human pose from RGB videos
- Solved issues such as occlusion, domain-gap and temporal jitter leading to realistic and smooth 3D sequence reconstructions on multiple in-the-wild video datasets

Learning to Generate Synthetic Data via Compositing

May 2018 – Nov 2018

Advisors: James Rehg, Amit Agrawal, Amrith Tyagi

Amazon Lab126

- Proposed a network for generating novel composite images that retain scene context and realism
- Developed algorithms for efficient training of object detection and image classification models on synthetic composite data, using an online hard-positive mining approach
- Improved baseline Faster-RCNN mAP by 3.5% and baseline SSD mAP by 2.7% on various datasets.

Deep Spectral-based Shape Features for Alzheimer's Disease Classification

Feb 2016 – Jul 2016

Undergraduate Thesis, Advisor: Samuel Kadoury

Univ. of Montreal

- Developed an unsupervised framework for classification of Alzheimer's disease patients using noisy T1-weighted MRI brain images
- Proposed a combination of grey-matter voxel-based intensity variations and 3D structural (shape) features parameterized with a spherical-harmonics representation
- Results presented near state-of-the-art accuracies (>89%) – outperformed conventional MRI shape-based strategies by 22%-27%

C2F: Coarse-to-Fine Vision Control System for Automated Microassembly

May 2014 – Dec 2014

Advisor: H D Sharma

Central Electronics Engineering Research Institute, Pilani

- Developed a completely automated, visual-servoing based closed loop system to perform 3D micromanipulation and microassembly tasks
- Results led to a ~75% reduction in setup and run time as compared to manual operation, while mitigating any risk of collision during grasp-and-drop experiments

SCHOLARSHIPS AND AWARDS	• Winner of the MaX!mize startup incubation grant for agentic game development (50k €)	2016
	• Selected for the Doctoral Consortium at CVPR 2025 (35 accepted participants)	2025
	• Outstanding Reviewer Award – ECCV 2020, CVPR 2025, CVPR 2026	
	• Winner of the Meta Research PhD Fellowship Award 2023 (21/3200 applicants)	2023
	• Best business model and best pitch, Cyber Valley Startup Incubation Program 2022, Germany for our startup “YOGI – a virtual yoga classroom”	2022
	• IISc Bangalore Summer Research Fellowship – top 20 across India	2015
	• <i>Best Technical Association Award</i> , BITS-Pilani	2014
	• Tournament Winner, Cricket, Arena'13 National Sports Festival	2013
	• Undergraduate MERIT scholarship, BITS Pilani – top 2% students	2012
	• Founder President's Scholarship, Amity International – top student for 6 years	2011
	• Junior Science Talent Search Examination (JSTSE) Scholarship – Ranked 22 in 20,000 applicants	2008
ACADEMIC DUTIES	Organizer – RHOBIN Workshop, CVPR 2024, CVPR 2025	Website link
	• 36 participating teams and 142 submissions. Top 7 methods established new state-of-the-art [CVPR 2024]	
	Organizer – 3D HUMANS Workshop, CVPR (2024, 2025), ECCV 2026	Website link
	Reviewer – CVPR (2026, 2025, 2024, 2022, 2021, 2020), ICCV (2023, 2021), ECCV (2020, 2026), BMVC 2022, PG 2024	
	Session Chair – International Max Planck Research Schools (IMPRS) Interview Symposium, 2024	
TALKS	• <i>PICO: Reconstructing 3D People in Contact with Objects</i>	19 Oct 2025
	Oral presentation at ICCV 2025 (gDT-IV , Rhobin)	
	• <i>Physics-informed Modelling of Dynamic Humans and their Interactions</i>	28 Mar 2024
	AI Talks, National Technological University (NTU), Singapore	
	• <i>DECO: Dense Estimation of 3D Human-Scene Contact in the Wild</i>	4 Oct 2023
	Oral presentation at ICCV 2023	
PROFESSIONAL EXPERIENCE	Max Planck Institute for Intelligent Systems	Tübingen, Germany
	Postdoctoral researcher	Jan 2026 – present
	Meta	Zurich, Switzerland
	Research Intern	Oct 2024 – Mar 2025
	Built a multi-modal large foundation model (VLM) for grounding human activities within 3D scene and enabling advanced spatial reasoning queries using egocentric inputs from a head-mounted device	
	Epic Games	San Francisco, USA
	Research Intern	June 2023 – Dec 2023
	Build a novel motion model, HUMOS, capable of generating natural, physically-plausible and dynamically stable human motions given a target body shape	
	Amazon	Sunnyvale, USA
	Applied Scientist II (AS-II) (<i>promoted from AS-I in Sep 2020</i>)	Feb 2019 – Feb 2021
	Improved 3D human activity reconstruction from 2D videos for enhancing action recognition/detection. Supported Computer Vision algorithm development for the new Echo Show. Worked on virtual try-on and body measurement estimation from images.	
	Amazon Lab126	Cupertino, USA
	Applied Scientist Intern	May 2018 – Aug 2018
	Worked on task-aware generation of synthetic image composites for training deep networks	
	Franklin Templeton Investments	Hyderabad, India
	Summer Intern Project: Financial Modelling for Tactical Asset Allocation	May 2015 – Aug 2015
	Built machine-learning models for capturing statistical associations like lead-lag correlation and one directional causality which achieved a 12% improvement in hit-rate for forecasting yield-spreads (US-OAS)	
TEACHING EXPERIENCE	Teaching Assistant – 16-720: Computer Vision, Prof. Kris Kitani	Fall 2018, CMU
	Head Teaching Assistant – 16-385: Computer Vision, Prof. Ioannis Gkioulekas	Summer 2018, CMU
MENTORING	Alpar Cseke	Masters at University of Tübingen → Research Engineer at Meshcapade (2023 – Now)
	Agniv Chatterjee	Bachelors at Jadavpur University → Ph.D. at UT Austin (Georgios Pavlakos) (2022 – 2023)
ACADEMIC PROJECTS	Learning Scene Saliency Maps Using Superpixel-augmented Convolutional Neural Networks	Aug 2017 – Dec 2017
	• Extracted SLIC superpixel segmentations as input to a Siamese CNN, achieving 4x faster training times with improved spatial context and accuracy in output saliency maps	
	Towards Integrating Model Dynamics for Sample Efficient Reinforcement Learning	Jan 2017 – May 2017
	• Learned a dynamics model of the world by assuming domain-specific priors on real-world episodes. Used the learned dynamics model to augment real-world episodes as the training progressed	

- Established that augmenting real-world data using an approximate world-model tends to be significantly more sample efficient than naïve model-free reinforcement learning

LEADERSHIP

- Member, External Affairs Committee (Graduate Student Assembly), CMU
- Secretary, Electrical and Electronics Association, BITS Pilani
Led a team of 37 members. Organised 25 major events, 6 during the technical festival
- Computer Vision Mentor, Student Mentorship Program (SMP), BITS Pilani
Conducted evening classes for teaching 30 junior batch students
- Represented BITS Pilani cricket team in inter-college cricket tournaments and sports festivals
- Organizer of National Seminar on Indian Space Technology (NSIST-2014)

EXTRA-CURRICULAR

- Teaching volunteer at Nirmaan – BITS Pilani | www.nirmaan.org Mar 2014 – Dec 2015
- Teaching volunteer at LaSalle Boys and Girls Club, Montreal | www.bgclasalle.com Mar 2016 – Jul 2016
- Teaching volunteer at Amitasha – Teaching the girl child | www.amity.edu/amitasha Mar 2009 – Mar 2010